

AI-Claim Audit for Robust Vision

MILS Assignment Report

1. 實驗設定 (Problem Setting)

本次實驗旨在比較不同模型架構（高性能大型架構、輕量化架構、以及不同訓練方式的 Transformer）在面對分佈偏移（Distribution Shift）時的穩健性指標。

- 基準資料集 (**Clean**): ImageNet-V2
- 偏移資料集 (**Shifted**): ImageNet-R (包含 16 種風格變體)
- 模型列表: ResNeXt-101, MobileNet-V3, ViT-B16-V1, ViT-B16-SWAG

2. 實驗環境 (Experimental Environment)

項目	內容
硬體	NVIDIA GeForce RTX 5090 (32GB VRAM)
軟體	PyTorch 2.11.0, CUDA 12.8

3. 穩健性指標評估 (Robustness Metrics)

指標 (Metrics)	ResNeXt-101	MobileNet-V3	ViT-B16-V1	ViT-B16-SWAG
Clean Accuracy	85.83%	79.25%	85.00%	87.25%
Shifted Accuracy	31.47%	20.60%	28.07%	50.30%
Accuracy Drop	0.5436	0.5865	0.5693	0.3695
Relative Drop	63.33%	74.01%	66.97%	42.35%
Avg Robust Acc	29.48%	18.62%	26.13%	46.78%
Worst-case Acc	17.03%	7.92%	11.99%	36.22%
Wrong Conf	23.47%	24.10%	41.03%	45.74%
Rej Robust (80%)	36.34%	24.56%	33.81%	59.06%

4. AI Claim Audit (假說審核)

* 審核基於 ResNeXt-101 與 MobileNet-V3 之對比。

ID	假說內容 (Claim)	證據 (Evidence)	審核結果
C1	ResNeXt-101 has a smaller average Accuracy Drop on ImageNet-R than MobileNet-V3.	ResNeXt-101 Drop: 0.5436, MobileNet-V3 Drop: 0.5865	Supported
C2	ResNeXt-101's Accuracy Drop is largest on 'cartoon' and 'sketch' sub-categories.	Max ResNeXt Drop on graffiti (0.6881).	Refuted

C3	Wrong Confidence is higher for MobileNet-V3 than ResNeXt-101.	ResNeXt-101 Wrong Conf: 0.2347, MobileNet-V3 Wrong Conf: 0.2410	Supported
C4	Both models have their highest Accuracy Drop on 'sculpture' renditions.	ResNeXt Max: graffiti, MobileNet Max: tattoo	Refuted
C5	The failure overlap between ResNeXt-101 and MobileNet-V3 is less than 50%.	Overlap Ratio: 79.90%	Refuted

5. 深入分析與觀察 (In-depth Analysis)

實驗結果顯示，**ViT-B16-SWAG** 在所有模型中展現了最強的穩健性，其 Accuracy Drop 最低且在極端偏移下的準確度最高。這證明了大規模弱監督預訓練（SWAG）能顯著提升模型對分佈偏移的抵抗力。

相對地，MobileNet-V3 雖然在基礎準確度上不俗，但在面對風格偏移時表現最為脆弱，其 Worst-case Accuracy 指標顯示在某些風格下幾近失效。

6. 失敗案例分析 (Failure Analysis)

ResNeXt-101 Fail	MobileNet-V3 Fail	ViT-B16-V1 Fail
ViT-B16-SWAG Fail		

7. 結語 (Conclusion)

本實驗透過對比四種不同規模與訓練方式的模型，揭示了模型架構與預訓練策略對穩健性的決定性影響。SWAG 預訓練與大型架構（ResNeXt）在處理複雜風格偏移時具有明顯優勢。